

INDEX

1. Problem Description
 2. Architecture
 3. Models Used (Applied / Planned)
 4. Evaluation Metrics Used
 5. Preliminary Results and Visualizations
 6. References
-

1. Problem Description

Agriculture plays an important role in the economy and food production system of every country. Different agricultural regions have different environmental conditions such as rainfall, temperature, soil quality, irrigation facilities, and crop productivity. Manually analyzing large agricultural datasets to identify similar farming regions is difficult and time-consuming. Therefore, there is a need for an intelligent analytical system that can automatically group agricultural regions based on their similarities.

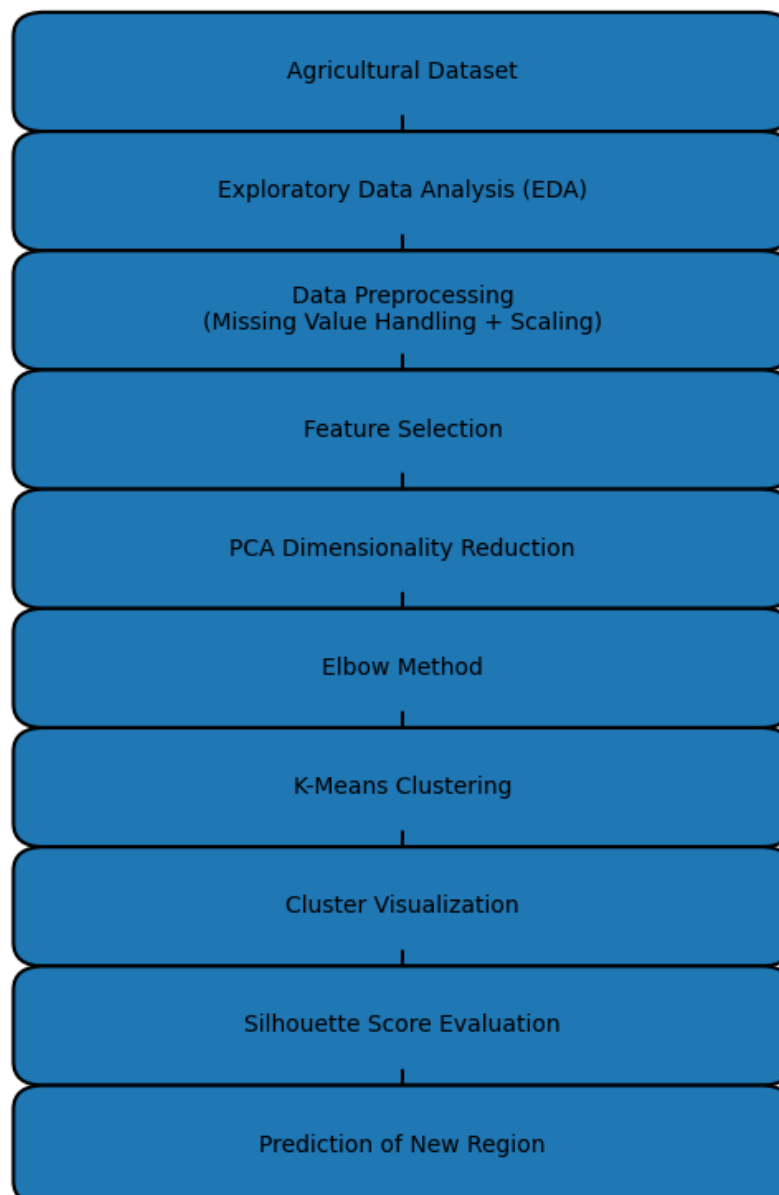
This project aims to develop a K-Means clustering-based analytical system to group agricultural regions using machine learning techniques. The dataset contains 20 region-level features including rainfall, temperature, soil moisture, soil pH, nitrogen content, phosphorus content, potassium content, irrigation coverage, crop area, fertilizer usage, pesticide usage, sunlight hours, humidity, altitude, wind speed, groundwater level, crop diversity index, farming mechanization score, historical yield variance, and average crop yield.

The project begins with Exploratory Data Analysis (EDA) to understand feature distributions, correlations, and hidden patterns in the dataset. Missing values are handled using imputation techniques, and feature normalization is applied to ensure all features are on a comparable scale. Feature selection methods are used to remove less relevant or highly correlated features.

The K-Means clustering algorithm is then applied to group agricultural regions into clusters with similar environmental and crop yield characteristics. The optimal number of clusters is determined using the Elbow Method and Silhouette Score. PCA (Principal Component Analysis) is also applied for dimensionality reduction and better visualization of clusters.

Finally, the trained model is used to predict the cluster label for new or unseen agricultural regional data, demonstrating the real-world applicability of the system. The project helps in identifying similar agricultural zones, which can support better farming strategies, crop planning, and resource management.

2. Architecture



3. Models Used (Applied / Planned)

1. K-Means Clustering Algorithm

K-Means is an unsupervised machine learning algorithm used to divide data into different groups or clusters based on similarities between data points. The algorithm works by selecting K cluster centroids and assigning each data point to the nearest centroid. The centroids are updated repeatedly until the clusters become stable.

In this project, K-Means clustering is used to group agricultural regions having similar environmental and crop yield conditions. The algorithm helps in identifying high-yield, medium-yield, and low-yield farming regions based on multiple agricultural parameters.

Characteristics of K-Means

Parameter	Description
Learning Type	Unsupervised Learning
Purpose	Group Similar Regions
Input	Numerical Agricultural Features
Output	Cluster Labels
Training Speed	Fast
Best For	Large Numerical Datasets

2. PCA (Principal Component Analysis)

PCA is a dimensionality reduction technique used to reduce the number of features while preserving important information. Since agricultural datasets contain many correlated features, PCA helps in simplifying the dataset and improving clustering performance.

PCA also helps in visualizing high-dimensional agricultural data in two-dimensional space for better understanding of clusters.

Characteristics of PCA

Parameter	Description
Type	Dimensionality Reduction
Purpose	Reduce Complexity
Output	Principal Components
Advantage	Better Visualization
Benefit	Improves Clustering Efficiency

3. Elbow Method

The Elbow Method is used to determine the optimal number of clusters (K) for the K-Means algorithm. It calculates the Within Cluster Sum of Squares (WCSS) for different K values and plots a graph. The point where the graph bends like an elbow is selected as the optimal K value.

4. Evaluation Metrics Used

1. Silhouette Score

Silhouette Score is used to evaluate the quality of clustering. It measures how similar a data point is to its own cluster compared to other clusters.

Interpretation:

- Close to +1 → Good Clustering
- Around 0 → Overlapping Clusters
- Negative → Poor Clustering

In this project, Silhouette Score is used instead of accuracy because K-Means is an unsupervised learning algorithm.

2. WCSS (Within Cluster Sum of Squares)

WCSS measures the compactness of clusters. Lower WCSS values indicate better clustering performance. WCSS values are used in the Elbow Method to identify the optimal number of clusters.

3. Cluster Visualization

Cluster visualization helps in understanding how agricultural regions are grouped. PCA-based scatter plots are used to visually interpret cluster formation.

5. Preliminary Results and Visualizations

Fig 1. Correlation Heatmap

The heatmap shows relationships between agricultural features such as rainfall, temperature, soil nutrients, and crop yield. Strong positive and negative correlations help identify important features for clustering.

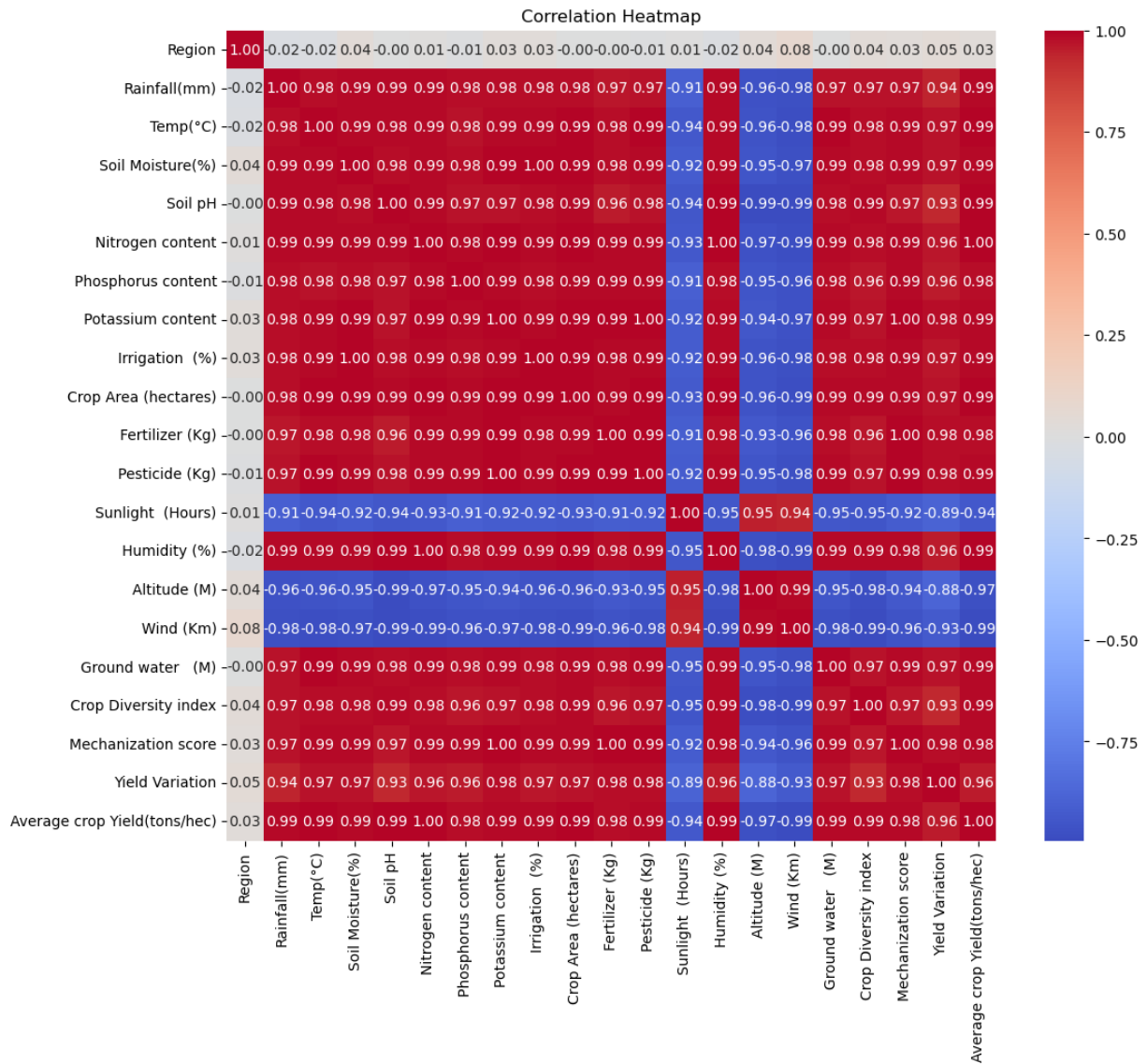


Fig 2. Elbow Method Graph

The Elbow Method graph is used to determine the optimal number of clusters by analyzing WCSS values for different K values.

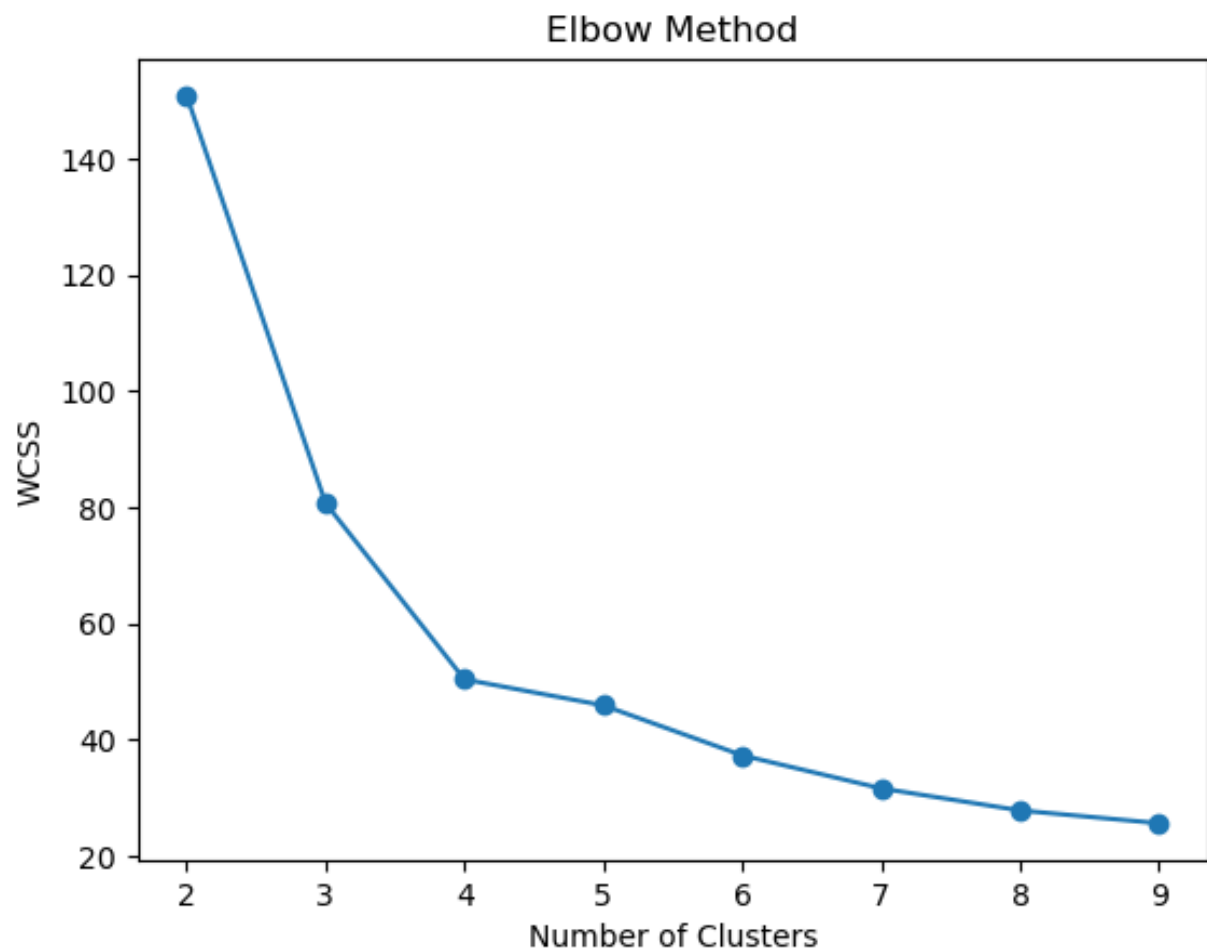


Fig 3. PCA Visualization

PCA reduces the dataset into two dimensions and helps visualize agricultural regions in a feature space before clustering.

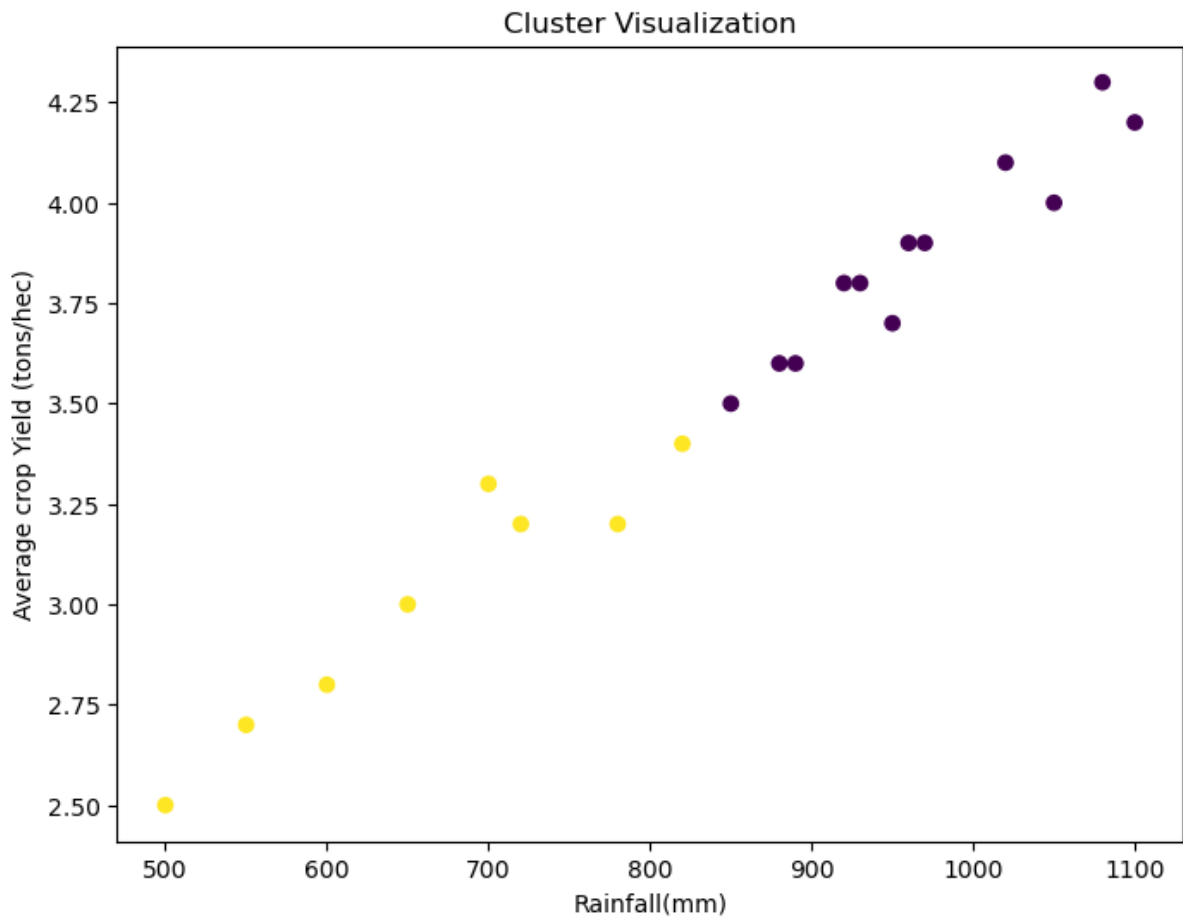


Fig 4. Final K-Means Clusters

The final scatter plot displays grouped agricultural regions based on environmental and crop yield similarities.

	Region	Rainfall(mm)	Temp(°C)	Soil Moisture(%)	Soil pH \
Pesticide (Kg)					
14	9.0	500.000000	19.0	30.000000	5.500000
15	11.5	575.000000	20.0	34.000000	5.750000
17	14.0	650.000000	21.0	38.000000	6.000000
18	9.5	750.000000	22.0	40.500000	6.200000
19	9.5	760.000000	23.0	43.000000	6.350000
20	10.0	873.333333	24.0	46.666667	6.566667
21	10.5	940.000000	25.0	51.500000	6.700000
22	11.0	950.000000	26.0	52.333333	6.800000
24	12.5	1035.000000	27.0	57.500000	6.900000
25	4.0	1100.000000	28.0	60.000000	7.000000
26	13.0	1080.000000	28.0	61.000000	7.100000

	Nitrogen content	Phosphorus content	Potassium content \
Pesticide (Kg)			
14	55.000000	28.000000	35.000000
15	59.000000	29.500000	37.500000
17	62.000000	33.000000	40.000000
18	65.000000	37.500000	42.000000
19	67.000000	37.500000	43.500000
20	71.000000	40.333333	45.666667
21	72.500000	41.500000	47.500000
22	74.666667	42.666667	49.000000
24	78.500000	47.500000	52.500000
25	80.000000	50.000000	55.000000
26	82.000000	51.000000	56.000000

	Irrigation (%)	Crop Area (hectares)	Fertilizer (Kg) \
Pesticide (Kg)			
14	45.000000	800.000000	120.000000
15	48.500000	875.000000	127.500000
17	52.000000	1000.000000	135.000000
18	56.000000	1110.000000	141.000000
19	59.000000	1165.000000	146.500000
20	61.666667	1236.666667	150.666667
21	67.000000	1360.000000	156.000000
22	66.666667	1390.000000	159.333333
24	72.500000	1510.000000	175.500000
25	75.000000	1600.000000	180.000000
26	76.000000	1620.000000	182.000000

	Sunlight (Hours)	Humidity (%)	Altitude (M)	Wind (Km) \
Pesticide (Kg)				
14	10.0	50.000000	600.000000	18.000000
15	9.5	54.000000	525.000000	16.500000
17	9.0	58.000000	450.000000	15.000000
18	8.5	60.500000	380.000000	14.000000
19	8.0	63.000000	335.000000	13.000000
20	8.0	65.666667	296.666667	12.000000
21	8.0	67.500000	272.500000	11.000000
22	7.0	69.666667	256.666667	10.666667
24	7.0	72.500000	225.000000	10.000000
25	7.0	75.000000	200.000000	9.000000

26 7.0 76.000000 210.000000 9.000000

	Ground water (M)	Crop Diversity index	Mechanization score \
Pesticide (Kg)			
14	8.000000	0.300000	40.0
15	9.500000	0.375000	43.5
17	11.000000	0.500000	48.0
18	12.500000	0.525000	50.5
19	13.500000	0.600000	53.0
20	14.666667	0.633333	56.0
21	16.000000	0.700000	58.5
22	17.666667	0.713333	61.0
24	19.000000	0.765000	68.5
25	20.000000	0.800000	70.0
26	21.000000	0.820000	72.0

	Yield Variation	Average crop Yield(tons/hect)
Pesticide (Kg)		
14	0.900000	2.500000
15	1.000000	2.750000
17	1.100000	3.000000
18	1.150000	3.200000
19	1.200000	3.350000
20	1.266667	3.566667
21	1.400000	3.750000
22	1.500000	3.866667
24	1.700000	4.050000
25	1.800000	4.200000
26	1.900000	4.300000

Fig 5. Silhouette Score Output

The Silhouette Score evaluates the quality and separation of clusters formed by the K-Means algorithm.

K=2, Silhouette Score=0.477
K=3, Silhouette Score=0.411
K=4, Silhouette Score=0.396
K=5, Silhouette Score=0.319
K=6, Silhouette Score=0.258
K=7, Silhouette Score=0.190
K=8, Silhouette Score=0.159
K=9, Silhouette Score=0.082
K=10, Silhouette Score=0.053

6. References

Books:

1. Artificial Intelligence and Machine Learning – S. Chand
2. Machine Learning with Python – O'Reilly Publication

Online Resources:

1. [Scikit-learn KMeans Documentation](#)
2. [Scikit-learn PCA Documentation](#)
3. [GeeksforGeeks K-Means Clustering](#)
4. https://www.geeksforgeeks.org/silhouette-index-cluster-validity-index-set-2-using-matlab/?utm_source=chatgpt.com